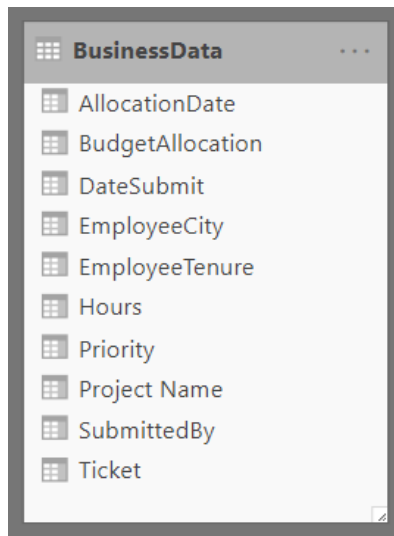
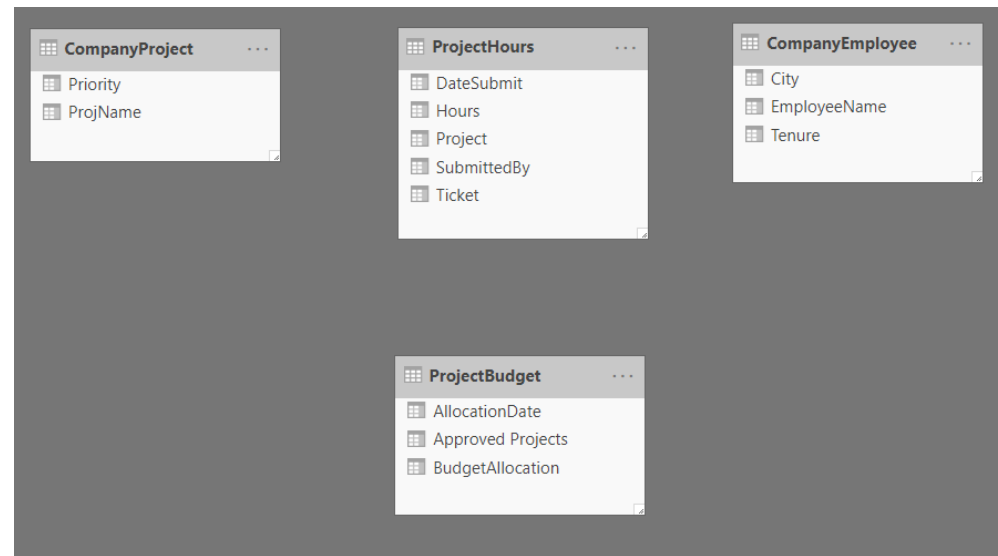

From Flat Tables to Data Model in Power BI

In this tutorial, you will learn how to create a Power BI model from a flat table with many columns. We will start from the Business Data table shown below on the left and we will split that into smaller tables (Fact Table and Look-up or Dimensions Tables) to create the Data Model on the right.



BusinessData	
AllocationDate	
BudgetAllocation	
DateSubmit	
EmployeeCity	
EmployeeTenure	
Hours	
Priority	
Project Name	
SubmittedBy	
Ticket	



CompanyProject	
Priority	
ProjName	

ProjectHours	
DateSubmit	
Hours	
Project	
SubmittedBy	
Ticket	

CompanyEmployee	
City	
EmployeeName	
Tenure	

ProjectBudget	
AllocationDate	
Approved Projects	
BudgetAllocation	

You can download the **BusinessData** Excel file from Blackboard. Click on Get Data from the Home ribbon, select Excel type and click Connect. Browse to the folder where you saved the BusinessData file and click Open. In the Navigator window, tick the BusinessData box and click Load (see below).

Navigator



Display Options ▾ 

- BusinessData.xlsx [1]
- ☒  BusinessData

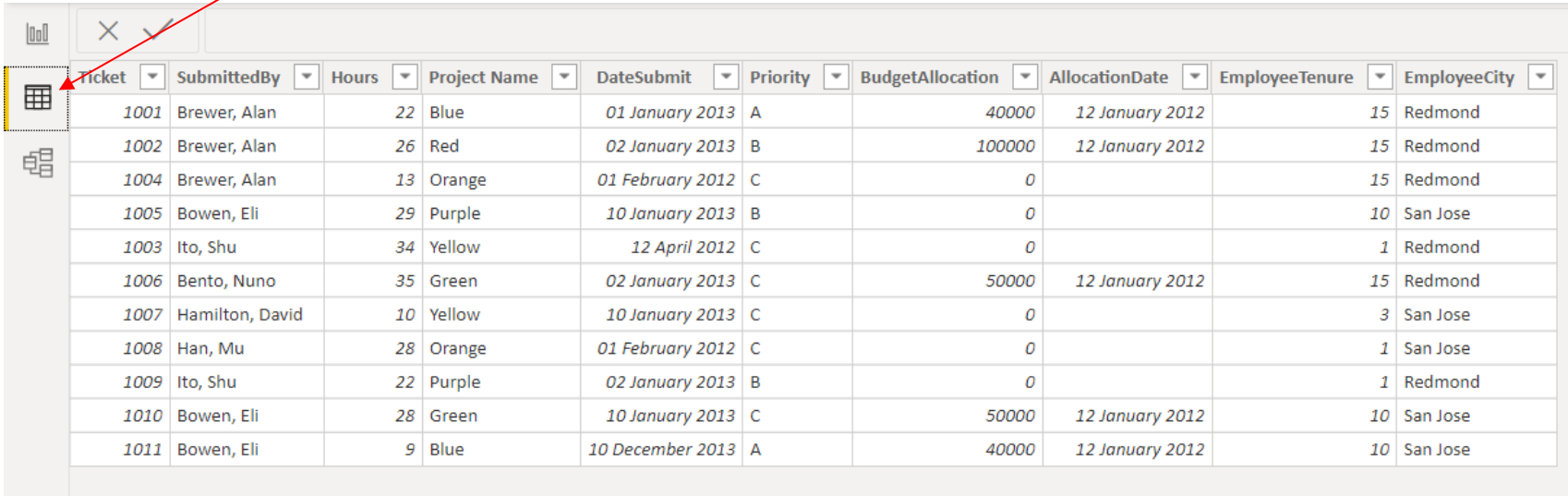
BusinessData

Ticket	SubmittedBy	Hours	Project Name	DateSubmit	Priority
1001	Brewer, Alan	22	Blue	01/01/2013	A
1002	Brewer, Alan	26	Red	02/01/2013	B
1004	Brewer, Alan	13	Orange	01/02/2012	C
1005	Bowen, Eli	29	Purple	10/01/2013	B
1003	Ito, Shu	34	Yellow	12/04/2012	C
1006	Bento, Nuno	35	Green	02/01/2013	C
1007	Hamilton, David	10	Yellow	10/01/2013	C
1008	Han, Mu	28	Orange	01/02/2012	C
1009	Ito, Shu	22	Purple	02/01/2013	B
1010	Bowen, Eli	28	Green	10/01/2013	C
1011	Bowen, Eli	9	Blue	10/12/2013	A



Load
Transform Data
Cancel

Click on the Data view on the left and let's have a look at the BusinessData table.



Ticket	SubmittedBy	Hours	Project Name	DateSubmit	Priority	BudgetAllocation	AllocationDate	EmployeeTenure	EmployeeCity
1001	Brewer, Alan	22	Blue	01 January 2013	A	40000	12 January 2012	15	Redmond
1002	Brewer, Alan	26	Red	02 January 2013	B	100000	12 January 2012	15	Redmond
1004	Brewer, Alan	13	Orange	01 February 2012	C	0		15	Redmond
1005	Bowen, Eli	29	Purple	10 January 2013	B	0		10	San Jose
1003	Ito, Shu	34	Yellow	12 April 2012	C	0		1	Redmond
1006	Bento, Nuno	35	Green	02 January 2013	C	50000	12 January 2012	15	Redmond
1007	Hamilton, David	10	Yellow	10 January 2013	C	0		3	San Jose
1008	Han, Mu	28	Orange	01 February 2012	C	0		1	San Jose
1009	Ito, Shu	22	Purple	02 January 2013	B	0		1	Redmond
1010	Bowen, Eli	28	Green	10 January 2013	C	50000	12 January 2012	10	San Jose
1011	Bowen, Eli	9	Blue	10 December 2013	A	40000	12 January 2012	10	San Jose

This table collects the data related to the projects approved in a company.

The **Ticket** column shows the Ticket's codes, you can think of this as the number of each "transaction".

The **SubmittedBy** column shows the name of the Employee that has submitted the request for approval for a specific project.

The **Hours** column shows how many hours the project requires.

The **ProjectName** column reports the name of the project (I have just used colours names to make this simpler).

The **DateSubmit** column shows the submission date of the project.

The **Priority** column reports the priority of each project, where A is the "high-priority", B is "medium-priority" and C is "low-priority".

The **BudgetAllocation** column reports the amount of money allocated for each project. Some projects do not have a Budget Allocation because they have not been approved yet.

The **AllocationDate** column reports the data when the project has been approved. Some projects do not have an Allocation Date because they have not been approved yet.

The **EmployeeTenure** column shows the amount of time each employee has been working for the company (in months).

The **EmployeeCity** column shows the city of each employee.

As you might have noticed, there is a lot of data redundancy in the table. There are many columns (EmployeeTenure and EmployeeCity for example) that have the same values repeated many times. Let's see how we can reduce data redundancy and create a well-structured data model.

Step 1

The first table we want to create is the **CompanyEmployee** table. This table will store the Employee Name, Tenure and City. Click on **Transform Data** (in the Home Ribbon) to open the Power Query Editor below.

Queries [1]    = Table.TransformColumnTypes("#Promoted Headers",{{"Ticket", Int64.Type}, {"SubmittedBy", type text}, {"Hours",

	 Ticket	 SubmittedBy	 Hours	 Project Name	 DateSubmit	 Priority
1	1001	Brewer, Alan	22	Blue	01/01/2013	A
2	1002	Brewer, Alan	26	Red	02/01/2013	B
3	1004	Brewer, Alan	13	Orange	01/02/2012	C
4	1005	Bowen, Eli	29	Purple	10/01/2013	B
5	1003	Ito, Shu	34	Yellow	12/04/2012	C
6	1006	Bento, Nuno	35	Green	02/01/2013	C
7	1007	Hamilton, David	10	Yellow	10/01/2013	C
8	1008	Han, Mu	28	Orange	01/02/2012	C
9	1009	Ito, Shu	22	Purple	02/01/2013	B
10	1010	Bowen, Eli	28	Green	10/01/2013	C
11	1011	Bowen, Eli	9	Blue	10/12/2013	A

To create the new table, follow the steps below (check the video-tutorial if you are unsure about any of the steps below: [link to video](#)):

- 1) Right-click on BusinessData table on the left and select "Duplicate"
- 2) A new table will appear called BusinessData(2)
- 3) Rename this table as CompanyEmployee by right-clicking on it and selecting Rename

- 4) Now let's remove all the columns that we don't need. To do so, click on the header of the SubmittedBy column, then press Ctrl and click on the EmployeeTenure and EmployeeCity columns. You should now have three columns selected.
- 5) Once the three columns are selected, right click on any of the headers of the three columns and select "Remove Other Columns".

Your table should now look like the one below.

Queries [2] <

BusinessData

CompanyEmployee

✕ ✓ fx = Table.SelectColumns("#Changed Type",{ "EmployeeCity", "EmployeeTenure", "SubmittedBy" })

	EmployeeCity	EmployeeTenure	SubmittedBy
1	Redmond	15	Brewer, Alan
2	Redmond	15	Brewer, Alan
3	Redmond	15	Brewer, Alan
4	San Jose	10	Bowen, Eli
5	Redmond	1	Ito, Shu
6	Redmond	15	Bento, Nuno
7	San Jose	3	Hamilton, David
8	San Jose	1	Han, Mu
9	Redmond	1	Ito, Shu
10	San Jose	10	Bowen, Eli
11	San Jose	10	Bowen, Eli

- 6) Rename the SubmittedBy column as "EmployeeName", by right-clicking on the column's header and selecting "Rename".
- 7) To move the EmployeeName column as the first column, click on the column header and drag-and-drop it on the left on the EmployeeCity column.
- 8) The final result is shown below.

Queries [2] ✕ ✓ fx = Table.ReorderColumns(#"Renamed Columns",{"EmployeeName", "Emp]

	A ^B EmployeeName	A ^B EmployeeCity	1 ² EmployeeTenure
1	Brewer, Alan	Redmond	15
2	Brewer, Alan	Redmond	15
3	Brewer, Alan	Redmond	15
4	Bowen, Eli	San Jose	10
5	Ito, Shu	Redmond	1
6	Bento, Nuno	Redmond	15
7	Hamilton, David	San Jose	3
8	Han, Mu	San Jose	1
9	Ito, Shu	Redmond	1
10	Bowen, Eli	San Jose	10
11	Bowen, Eli	San Jose	10

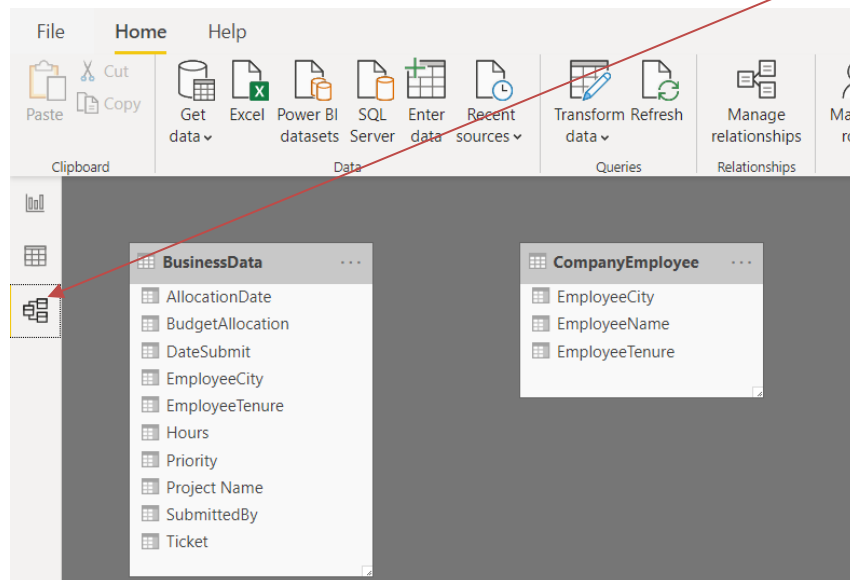
This table has many lines that are repeating. To remove them, right-click on the EmployeeName header and select Remove Duplicates (see below)

	A ^B EmployeeName	A ^B EmployeeCity	1 ² EmployeeTenure
1	Brewer, Alan		15
2	Brewer, Alan		15
3	Brewer, Alan		15
4	Bowen, Eli		10
5	Ito, Shu		1
6	Bento, Nuno		15
7	Hamilton, David		3
8	Han, Mu		1
9	Ito, Shu		1
10	Bowen, Eli		10
11	Bowen, Eli		10

The final table will look like this. We have already reduced data redundancy.

	A ^B _C EmployeeName	A ^B _C EmployeeCity	1 ² ₃ EmployeeTenure
1	Brewer, Alan	Redmond	15
2	Bowen, Eli	San Jose	10
3	Ito, Shu	Redmond	1
4	Bento, Nuno	Redmond	15
5	Hamilton, David	San Jose	3
6	Han, Mu	San Jose	1

Click on Close&Apply to apply these changes and select the Model tab. You should now see the two tables below.



The **CompanyEmployee** table stores information of the company employees.

Step 2

The next step is to create the **ProjectHours** table. This will store information about the Tickets code (Ticket), SubmittedBy, Hours of the project, ProjectName and the submission date (DateSubmit).

As done before, click on Transform Data in the Home ribbon to open the Query Editor and follow the steps below.

- 1) Replicate the BusinessData table, by right-clicking on it and selecting "Duplicate"
- 2) This will create a new table called BusinessData(2). Right-click on it and select "Rename" to change the name into ProjectHours
- 3) Now let's select the columns that we want to keep in this table. Click on the headers of the following columns and press Ctrl in your keyboard for multi-selection: Ticket, SubmittedBy, Hours, ProjectName and DateSubmit.
- 4) Once all the five columns are highlighted, right-click on the column header of any of them and select "Remove Other Columns".
- 5) Rename the SubmittedBy Column as "EmployeeName". To do so, right-click on that column and select "Rename".
- 6) This will leave only the columns below:

Queries [3] <

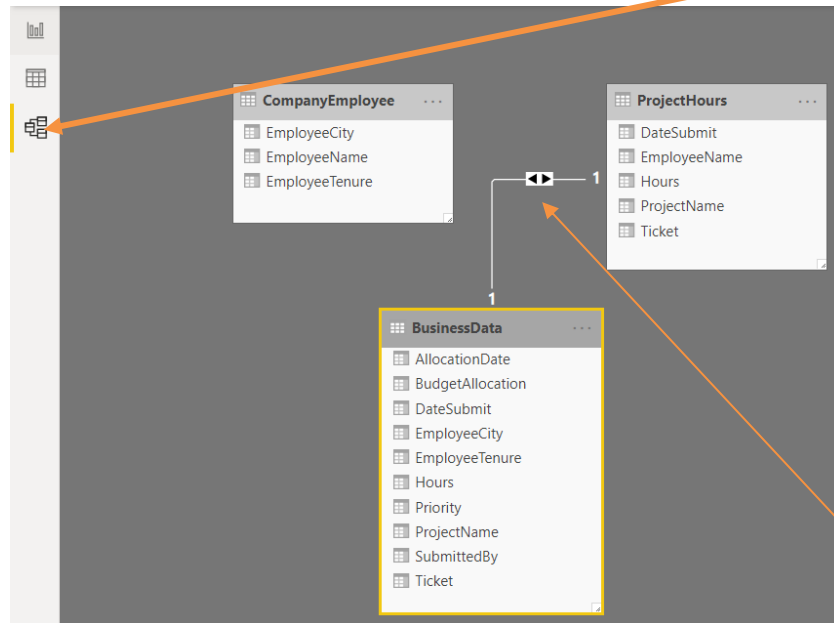
BusinessData
CompanyEmployee
ProjectHours

fx = Table.RenameColumns(#"Removed Other Columns",{{"SubmittedBy", "EmployeeName"}})

	Ticket	EmployeeName	Hours	ProjectName	DateSubmit
1	1001	Brewer, Alan	22	Blue	01/01/2013
2	1002	Brewer, Alan	26	Red	02/01/2013
3	1004	Brewer, Alan	13	Orange	01/02/2012
4	1005	Bowen, Eli	29	Purple	10/01/2013
5	1003	Ito, Shu	34	Yellow	12/04/2012
6	1006	Bento, Nuno	35	Green	02/01/2013
7	1007	Hamilton, David	10	Yellow	10/01/2013
8	1008	Han, Mu	28	Orange	01/02/2012
9	1009	Ito, Shu	22	Purple	02/01/2013
10	1010	Bowen, Eli	28	Green	10/01/2013
11	1011	Bowen, Eli	9	Blue	10/12/2013

You should notice that all the lines in this table are different. So there is no data redundancy here.

Click on Close&Apply to apply your changes and select the Model tab. You should now see the three tables below.



You now have three tables in your model. Power BI has created a link (relationships) for you between two tables. Don't worry about this for now. In this tutorial, we are focusing on splitting the big table into smaller ones. We will deal with relationships in the next lesson.

If you don't have that link, don't worry. We will not need it for now.

Notice that the two new tables **CompanyEmployee** and **ProjectHours** have the "EmployeeName" column in common (here they are displayed as rows, but they represent columns in the original tables). This is the "common denominator" than we will use to create a link between them.

If you don't see a double arrow here, that's OK. The new version of Power BI does not show it.

Step 3

The next step is create the **CompanyProject** table. This will contain the ProjectName column and the priority of the project (Priority column). We are going to repeat again the steps above.

- 1) Click on Transform Data in the Home ribbon to open the Query Editor.
- 2) Right-click on the BusinessData table on the left and select "Duplicate" to create a copy of this table.
- 3) This will create a new table called BusinessData (2). Right-click on it and select "Rename" to rename it as "CompanyProject"
- 4) Now, in the CompanyProject table, select the two columns that we want to keep: ProjectName and Priority. To do so, click on the column header of the ProjectName column, then press Ctrl on your keyboard and click on the column header of the Priority column.

- 5) You should now have two columns highlighted. Right-click on the header of any of these two columns and select "Remove Other Columns".
- 6) The final table should look like the one below:

Queries [4] <

BusinessData
CompanyEmployee
ProjectHours
CompanyProject

fx = Table.SelectColumns("#Renamed Columns",{

	ProjectName	Priority
1	Blue	A
2	Red	B
3	Orange	C
4	Purple	B
5	Yellow	C
6	Green	C
7	Yellow	C
8	Orange	C
9	Purple	B
10	Green	C
11	Blue	A

I am sure you have realized that in this table there are many lines repeating.

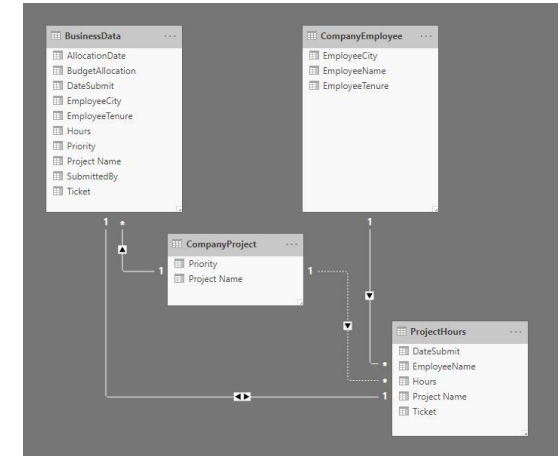
- 7) To delete the repeating lines, right-click on the ProjectName column's header and select "Remove Duplicates". Your new table will look like the one below:

	ProjectName	Priority
1	Blue	A
2	Red	B
3	Orange	C
4	Purple	B
5	Yellow	C
6	Green	C

Click on Close&Apply to apply these changes and select the Model tab. You should now see the four tables below.



Depending on your default settings, your model might look like the one on the right. If this is the case, don't worry, we will deal with these relationships later.



Notice that the table **CompanyProject** and **ProjectHours** have the "ProjectName" column in common (here they are displayed as rows, but they represent columns in the original tables). This is the "common denominator" than we will use to create a link between them.

Having a column in common between tables is really important when doing data modelling. Those columns in common will be then used to link tables and create reports.

So far, we have created three tables to store information about the Projects (CompanyProject table), the hours and submission dates (ProjectHours table) and details about the Employees (CompanyEmployees table). These tables are the "descriptive tables" (dimension tables). The next step is to create a table that stores information about any new projects approved. This table will be the Fact table since it is the only one that grows every time a new project is approved.

If you look at the BusinessData table above, you can see that all the variables in that table have been included in the three tables created so far. The only features missing are the AllocationDate and BudgetAllocation. We are going to add these two columns to the new table and we will also include the ProjectName column to make sure we will then be able to connect this table with the other tables. The ProjectName column will be the "common denominator" between this new table (the fact table) and the old tables (the dimension tables).

Step 4

To create the last table, we are going to repeat the same steps done before.

- 1) Click on Transform Data in the Home ribbon to open the Query Editor.
- 2) Right-click on the BusinessData table on the left and select "Duplicate" to create a copy of this table.
- 3) This will create a new table called BusinessData (2). Right-click on it and select "Rename" to rename it as "ProjectBudget"
- 4) Now, in the ProjectBudget table, select the three columns that we want to keep: ProjectName, BudgetAllocation and AllocationDate. To do so, click on the column header of the ProjectName column, then press Ctrl on your keyboard and click on the column header of the other two columns.
- 5) You should now have three columns highlighted. Right-click on the header of any of these three columns and select "Remove Other Columns".
- 6) The final table should look like the one below:

	A ^B C ProjectName	1 ² 3 BudgetAllocation	AllocationDate
1	Blue	40000	12/01/2012
2	Red	100000	12/01/2012
3	Orange	0	null
4	Purple	0	null
5	Yellow	0	null
6	Green	50000	12/01/2012
7	Yellow	0	null
8	Orange	0	null
9	Purple	0	null
10	Green	50000	12/01/2012
11	Blue	40000	12/01/2012

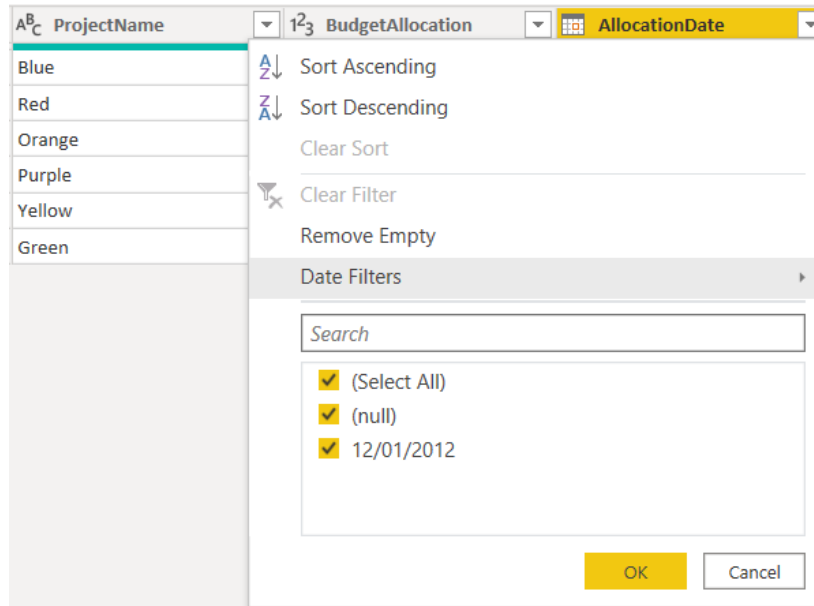
The null values tell you that there are some projects that have not been approved yet. For this reason, there is no BudgetAllocation or AllocationDate for them.

Notice that the Blue line is repeating twice. So let's see how we can remove duplicates.

- 7) As seen before, right click on the header of the ProjectName column and select "Remove Duplicates". You should now see the table below:

	ProjectName	BudgetAllocation	AllocationDate
1	Blue	40000	12/01/2012
2	Red	100000	12/01/2012
3	Orange	0	null
4	Purple	0	null
5	Yellow	0	null
6	Green	50000	12/01/2012

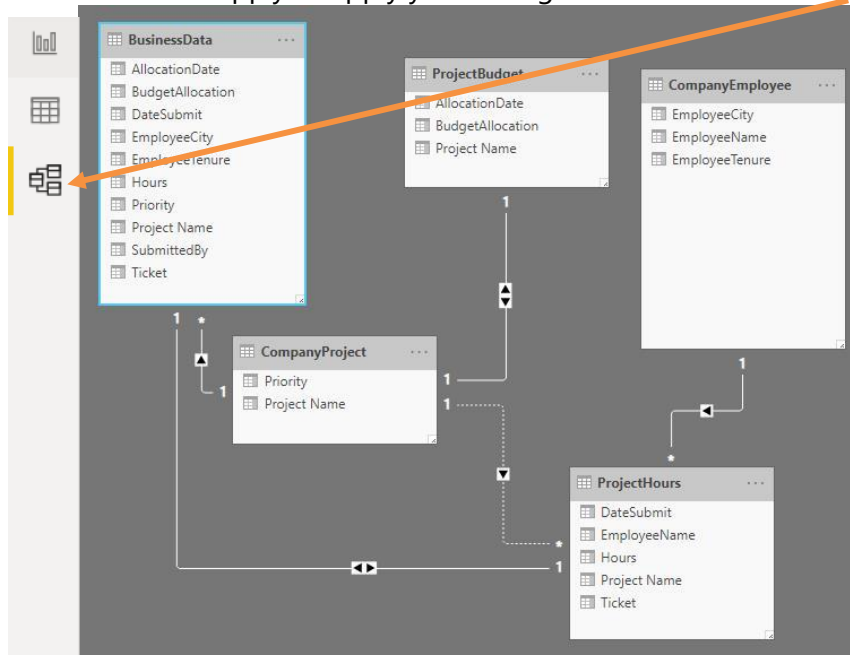
- 8) The next step is to remove the rows containing the "null" values. To do so, click on the drop down arrow in the AllocationDate column's header



and **untick the (null) cell** to remove the null values. Then click OK to close this window. The final table is displayed below

	ProjectName	BudgetAllocation	AllocationDate
1	Blue	40000	12/01/2012
2	Red	100000	12/01/2012
3	Green	50000	12/01/2012

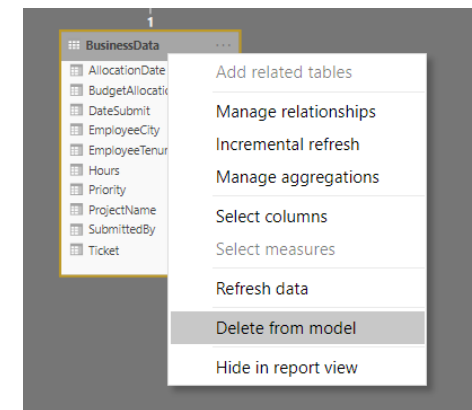
Click on Close&Apply to apply your changes and select the Model tab. You should now see the five tables below.



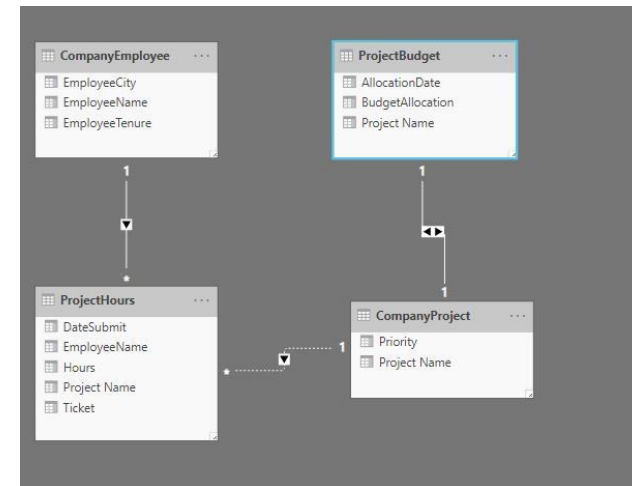
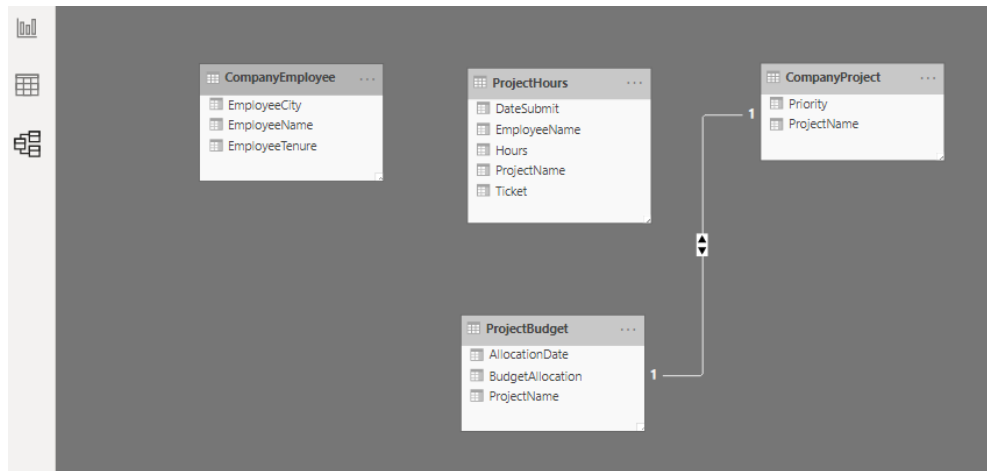
If you have different links between tables, that's fine for now.

Power BI has created a new link (relationship) for you. This is because Power BI has identified that the column "ProjectName" is present in both tables. If you can't see that link don't worry for now.

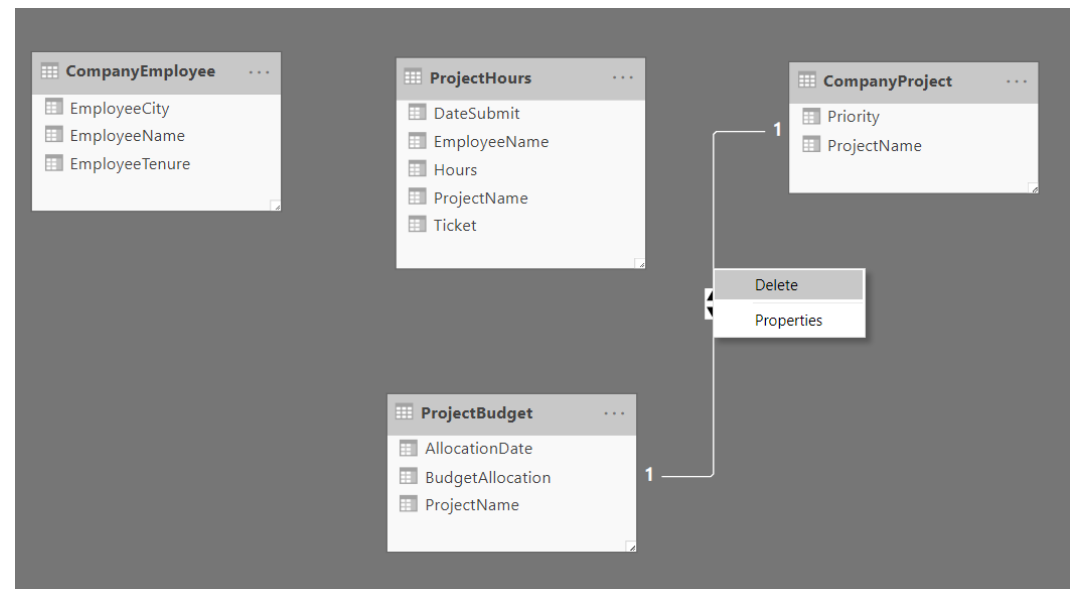
Now that we have split the BusinessData big table into smaller ones, we can remove that table. From the Model view, right-click on the BusinessData's header and select "Delete from model" (see image on the right). If a pop-up window will appear, click Delete.



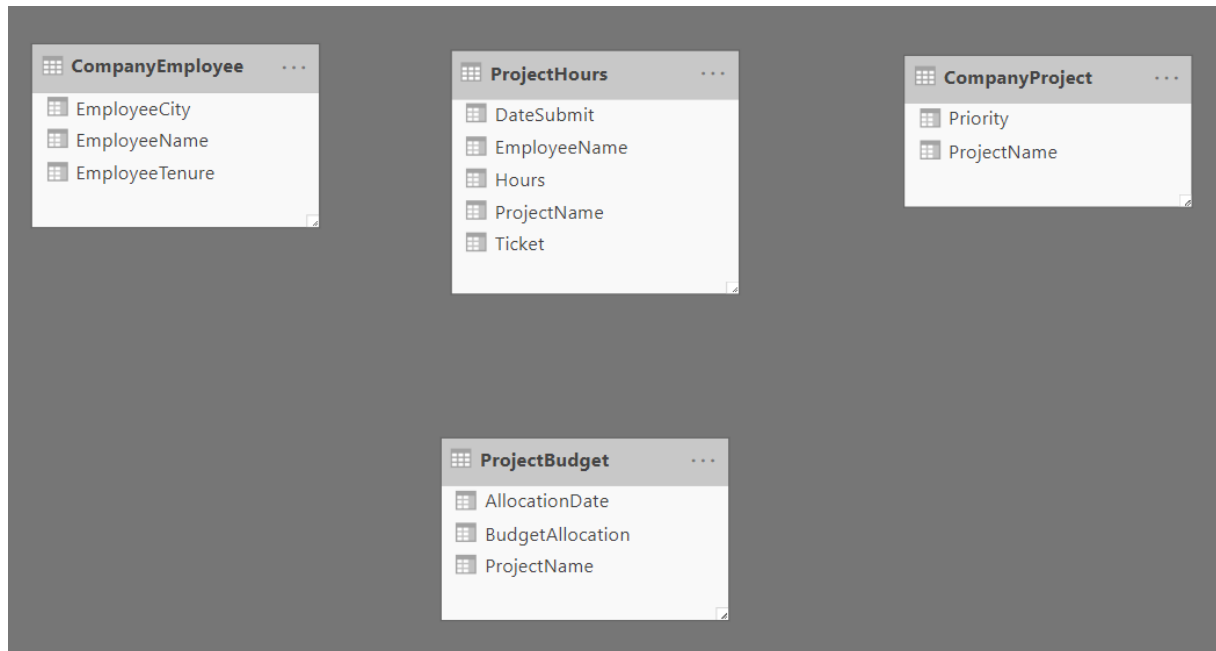
The new model will now look like the one below (you can rearrange the tables by clicking on the columns' headers to drag and drop them where you want). Note that you might have one or more relationships depending on your Power BI default options (see the two possible models below), before completing this tutorial, let's remove those links.



To remove a relationship, right-click on the link and select Delete (see below). In the pop-up window that asks if you want to delete the relationship, select Delete. If your model has other links, repeat this step to delete them all.



Your well-structured model is now ready. The fact table (ProjectBudget) is at the bottom and the Dimension tables are at the top.



In the next tutorial, you will learn how to connect the tables to create meaningful reports.

Don't forget to save this Power BI project as "BusinessData Model", you will need this in the next lesson.